CDT204 – Computer Architecture

> Date: Jan. 4th 2024
>
> Time: 8:30 – 12:30
>
> Help: Small calculator is allowed in the exam

The exam has 34 points. And 1 BONUS points. The grades will be awarded as follows:

- 3 : 17 Points
- 4 : 27 Points
- 5 : 31 Points

Important Notes:

- Give as full an answer as possible to obtain full marks. All calculations, approximations, assumptions and justifications must be reported for full credit unless stated otherwise. Please use figures and examples to clarify.
- If you do not understand a question clearly, you are allowed to call the teacher and ask.
- Write the question and part number on each page clearly.
- Answer each question on a separate page.
- In case you might have forgotten:

  1GB = 2^{10}MB = 2^{20}KB = 2^{30}B

  1 sec = 10^3 ms = 10^6 μs = 10^9 ns = 10^{12} ps

  $$S = \frac{T_0}{\frac{T_A}{F} + (T_0 - T_A)}$$

Live long and prosper

Task 1 (4p + 1p BONUS)

Consider the following binary representation: 1101 0110

a) What is the hexadecimal representation of that number? (1p)
b) What is its value in decimal form (base 10) if it contains an unsigned integer? (1p)
c) What is its value in decimal form (base 10) if it contains a two's complement signed integer? (1p)

Remember to write down your calculations and explain your solution. Writing down just the final answer will give you ZERO points, even if correct.

BONUS: You will get an extra point if you solve part a without converting the binary to decimal first (1p)

Task 2 (8p)

When a program is adapted to run on multiple processors in a multiprocessor system, the execution time on each processor is comprised of computing time and the overhead time required for locked critical sections and/or to send data from one processor to another.

Assume a program requires $t = 100s$ of execution time on one processor. When run p processors, each processor requires t/p seconds, as well as an additional $4s$ of overhead, irrespective of the number of processors.

A. Compute the per-processor execution time for 2, 4, 8, 16, 32, 64, and 128 processors. For each case, list the corresponding speedup relative to a single processor and the ratio between actual speedup versus ideal speedup (speedup if there was no overhead). (4p)
B. Draw a chart with number of processors on x-axis and the speedup on the y-axis. Plot both actual speedup and ideal speedups that you calculated for the previous part into this graph. Discuss your findings. (4p)

Task 3 (8p)

A processor has 12 stages. Each stage completes in a single cycle. The pipeline stalls on a conditional branch until the condition of the branch is evaluated. However, you do not know at which stage the branch condition is evaluated.

A. A program with 1000 instructions completes after 2211 cycles on this processor. If 200 of those instructions are conditional branches, at the end of which stage the branch instructions are resolved? (Assume that the pipeline does not stall for any other reason than the conditional branches). (3p)
B. In a new higher-performance version of the processor, a mysterious branch prediction mechanism is implemented, while keeping the rest of the design exactly as before. This new design completes the following code in 118 cycles. Assuming that the pipeline does not stall on data dependency, what can be the mysterious branch prediction mechanism and how it is configured? (5p)

```
LI $s1, 0
LI $t1, 1
LI $t5, 5
LI $t3, 3

LOOP_1:
  BEQ    $s1, $t5, LAST
  ADD    $s1, $s1, $t1
  LI     $S2, 0
LOOP_2:
  BEQ    $s2, $t3, LOOP_1
  ADD    $s2, $s2, $t1
  J      LOOP_2

LAST:
  LI     $s1, 1
```

Task 4 (6p)

Consider the following C code. Assume that each element of the two-dimensional arrays A and B is a 64-bit integer. Assume that cache block size is 16 bytes.

```
for(i=0; i<1024; i++)
  for(j=0; j<1024; j+=2)
         A[i][j] = B[i][0] + A[j][i];
```

A. Explain temporal and spatial locality. (2p)
B. Which variable references exhibit temporal locality? (2p)
C. Which variable references exhibit spatial locality? (2p)

Task 5 (8p)

You have been asked to improve the performance of an application developed by another programmer. In this application, one of the most executed parts of the code is as follows.

```
bool shot_by_laser[entries];

for(int i=0; i< entries; i++){

  if(objects.isCharacter[i]==true && objects.isAlive[i]==true)
        shot_by_laser[i] = true;
  else
        shot_by_laser[i] = false;

}
```

Assume that:

- All the variables are four-byte long.
- The number of entries is 32768 (2^{15}).
- The processor has a direct-mapped cache of 8MB with a cache block size of 64 bytes.
- A cache hit takes one cycle and a cache miss takes 100 cycles for both load and store.
- The starting address of `objects.isCharacter`, `objects.isAlive`, and `shot_by_laser` are 0x5000000, 0x06000000, and 0x07000000 respectively.
- The execution time of non-memory instructions is zero.

A. How many cycles are required to run this code? (5p)
B. What suggestions you might have to improve this code? (3p)